

Probit Offset with a pace factor

probit family, scored against the incumbent Offset model on the same margins on 2015–2026, 5 seed(s), from unrecorded commit

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Model identity and provenance

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL Probit Offset with a pace factor probit	COMPARATOR the incumbent Offset model on the same margins	SEEDS 5 7, 17, 42, 123, 256
TEST YEARS 12 2015-2026	BOOSTING ROUNDS n/a	SOURCE COMMIT n/a clean

Config digest unrecorded, model view bagged_probit_offset_pace, artifacts read from /Users/jlcheung/PycharmProjects/hkjc-mod/results/probit_models.

ENVIRONMENT

Component	Version
numpy	2.5.2
pandas	3.0.5
python	3.12.11
xgboost	3.4.1

Deltas in this report use model_minus_market_log_loss: race_multinomial_log_loss, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Executive summary

MEAN DELTA -0.000785 negative is better	YEARS BETTER 10/11	WORST YEAR 0.001175
CALIBRATED DELTA -0.000341 both sides calibrated		

Probit Offset with a pace factor is better than its comparator by 0.000785 `race_multinomial_log_loss`, measured against the incumbent Offset model on the same margins on the evaluated races. The direction holds in 10 of 11 evaluated years. Every seed agrees on the sign of the improvement.

Comparator and evaluation population

Same comparator as the market-anchored probit, so the two are directly comparable to each other: the only difference between them is the shared pace factor in the covariance.

TEST RACES
8,675

YEARS
12
2015-2026

SEEDS
5

HONEST-SCALE YEARS
11
years whose utility scale was fitted rather than fallen back

Some years fell back rather than fitting their own scale

2015 did not produce an honest utility scale (`insufficient_history`). Those years are excluded from the headline delta and from the bankroll paths, and shown separately in the year-by-year table: a fallback scale is not a measurement of the candidate.

Overall probability performance

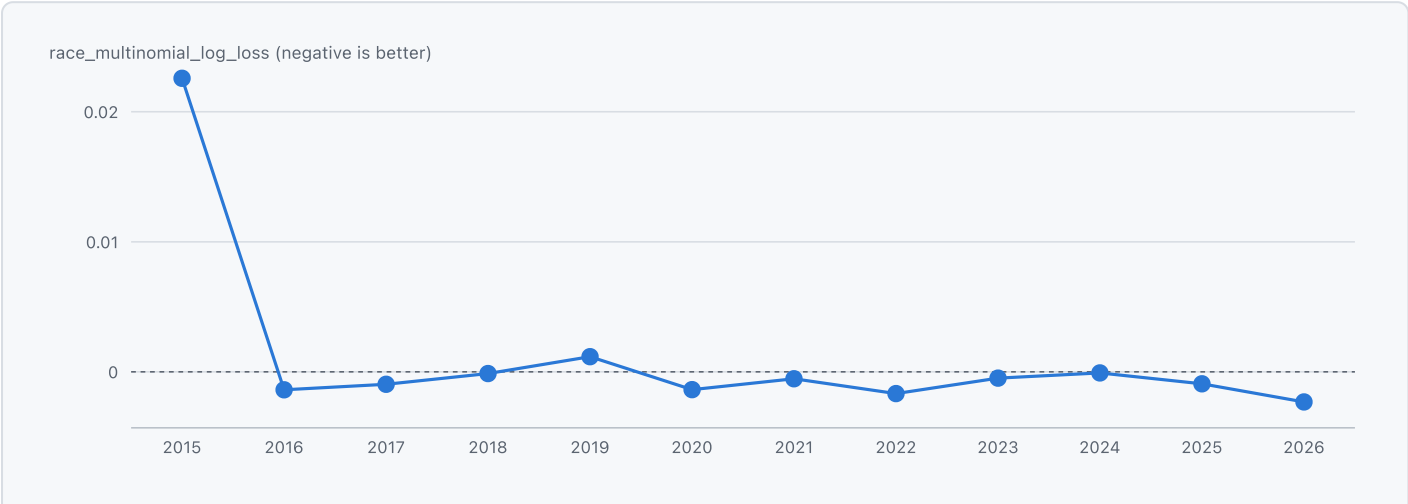
races	model log loss	comparator log loss	market log loss	honest delta	better year share
43,375	2.024605	2.025390	2.028805	-0.000785	0.909091

One row, pooled over every evaluated race. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	comparator log loss	market log loss	delta	honest
2015	3710	2.015068	1.992494	1.998720	0.022574	no
2016	3720	2.092451	2.093824	2.099277	-0.001373	yes
2017	3860	2.017101	2.018056	2.021559	-0.000955	yes
2018	3780	2.057580	2.057714	2.061504	-0.000134	yes
2019	3795	1.927395	1.926220	1.932827	0.001175	yes
2020	3995	2.006427	2.007792	2.010400	-0.001365	yes

test year	racess	model log loss	comparator log loss	market log loss	delta	honest
2021	3915	2.001404	2.001934	2.006659	-0.000530	yes
2022	3915	2.050094	2.051758	2.051195	-0.001663	yes
2023	3980	2.020665	2.021139	2.024066	-0.000475	yes
2024	3960	2.017508	2.017590	2.025744	-0.000083	yes
2025	3740	2.051190	2.052113	2.057397	-0.000923	yes
2026	1005	2.028838	2.031148	2.026224	-0.002310	yes



Mean delta by year Mean delta for each evaluated year. Zero is the comparator.

▼ Table view

Category	race_multinomial_log_loss (negative is better)
2015	0.022574
2016	-0.001373
2017	-0.000955
2018	-0.000134
2019	0.001175
2020	-0.001365
2021	-0.00053
2022	-0.001663
2023	-0.000475
2024	-0.000083
2025	-0.000923
2026	-0.00231

Ahead of its comparator in 10 of 12 years. 1 row(s) are italicised as partial or fallback years, shown rather than dropped so the evaluation population stays visible.

Calibration and sharpness

This architecture reports calibration through a paired comparison rather than binned reliability: candidate and comparator are each given a temperature on the same races, so the calibrated delta is the like-for-like figure. Fallback-scale years are excluded here, as they are from the headline.

test year	candidate calibrated loss	comparator calibrated loss	calibrated delta
2016	2.092447	2.093236	-0.000789
2017	2.018158	2.018225	-0.000067
2018	2.058258	2.057516	0.000742
2019	1.929684	1.927971	0.001712
2020	2.007505	2.009214	-0.001709
2021	2.000953	2.001587	-0.000633
2022	2.050874	2.051204	-0.000330
2023	2.021438	2.021954	-0.000517
2024	2.017259	2.017686	-0.000427
2025	2.053116	2.053894	-0.000779
2026	2.029934	2.030893	-0.000959

If the calibrated delta is materially smaller than the uncalibrated one, part of the candidate's advantage was a temperature the comparator could also have had.

Top-choice movement

Not applicable to this model. Top-choice movement compares a model's first pick against the market's. This candidate is scored against its own comparator on fixed mean utilities, not against the market, so a market top-choice comparison would answer a different question than the rest of this report.

Ensemble stability

Not applicable to this model. There is no ensemble to be stable. This architecture is a choice [link](#) applied to one set of mean utilities, not a bag of independently fitted members, so seed variation here is integration noise rather than model disagreement -- and that is reported in the appendix instead.

Feature and correction effects

Not applicable to this model. There are no per-feature effects to attribute. The [link](#) changes how utilities become probabilities; the utilities themselves come from the comparator's fit, whose feature effects are reported in that model's own report.

Representative races

Not applicable to this model. Representative races are chosen by how much a model's correction moved a runner's probability. This candidate applies a link to fixed mean utilities, so the movement to explain belongs to the comparator's fit and is shown in that model's report. A race chosen here would be chosen by integration noise.

Betting and bankroll analysis

Not applicable to this model. No bet ledger has been settled for Probit Offset with a pace factor. The probability comparison stands on its own; staking a candidate that has not passed its promotion gates would invite reading a bankroll path as a recommendation.

Simulation and uncertainty

MEAN DELTA -0.000677	INTERVAL -0.0012 to -0.0002 80% level	EXCLUDES ZERO FAVOURABLY yes
RACE DAYS 891 39,665 races	REPLICATES 2,000 seed 20260824	

A day-block bootstrap: race days are resampled within each test year, so the interval accounts for clustering within a meeting, and the interval lies entirely below zero. It conditions on the fitted model and does *not* include refit uncertainty.

What a simulation cannot tell you

The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

THE UTILITY SCALE

FITTED MINIMUM 1.9932	FITTED MAXIMUM 2.2280	THEORY 0.7797 the value theory predicts
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The fitted scale sits above the value theory predicts, in every year: 1.993 to 2.228 against 0.780 (the scale the fallback year used, which is the predeclared theoretical value). Theory gets this wrong and the fit is what the report uses -- a scale asserted from theory would divide the utilities by too little, and at that value the market-relative candidate loses instead of winning. A run that took the theoretical value on faith would have reported the wrong sign, which is why the first year -- which has no history and falls back to it -- is excluded from the headline and labelled.

CONFIGURATION

Item	Value
Architecture	probit_offset
Comparator field	offset
Integration	exact_gauss_hermite_quadrature
Shared factors	1
Factor loadings	['early_style_ewm']

PROMOTION GATES

Gate	Verdict
1. paired delta is negative	pass
2. interval excludes zero favourably	pass
3. no year degrades beyond tolerance	FAIL
4. survives nested calibration	pass
5. direction stable across seeds	pass
6. improvement exceeds integration error 10x	pass

5 of 6 checkable gate assertions hold. Passing them is not promotion. An interaction diagnostic across predeclared cohorts, reproduction from a clean identified commit, and confirmation on a future prospective period are not derivable from these artifacts. Production remains the five-seed Offset ensemble until they are met.

PER-SEED DELTA

Seed	Delta
7	-0.000809

Seed	Delta
17	-0.000759
42	-0.000705
123	-0.000778
256	-0.000875

5 of 5 seeds favour the candidate, and every seed agrees on the sign. Shown individually rather than only pooled, because a mean whose seeds disagree is a different claim from one whose seeds agree.

INTEGRATION SENSITIVITY

architecture	study	setting	value	metric	n races
probit_offset	utility_scale	2015	0.779697	n/a	0
probit_offset	factor_scale_grid	2016_scale_0.00	0	1.992242	0
probit_offset	factor_scale_grid	2016_scale_0.15	0.150000	1.992222	0
probit_offset	factor_scale_grid	2016_scale_0.30	0.300000	1.992200	0
probit_offset	factor_scale_grid	2016_scale_0.50	0.500000	1.992205	0
probit_offset	factor_scale_grid	2016_scale_0.80	0.800000	1.992289	0
probit_offset	utility_scale	2016	1.993223	n/a	0
probit_offset	factor_scale_grid	2017_scale_0.00	0	2.042575	0
probit_offset	factor_scale_grid	2017_scale_0.15	0.150000	2.042517	0
probit_offset	factor_scale_grid	2017_scale_0.30	0.300000	2.042405	0
probit_offset	factor_scale_grid	2017_scale_0.50	0.500000	2.042286	0
probit_offset	factor_scale_grid	2017_scale_0.80	0.800000	2.042203	0
probit_offset	utility_scale	2017	2.034700	n/a	0
probit_offset	factor_scale_grid	2018_scale_0.00	0	2.034032	0
probit_offset	factor_scale_grid	2018_scale_0.15	0.150000	2.033987	0
probit_offset	factor_scale_grid	2018_scale_0.30	0.300000	2.033898	0
probit_offset	factor_scale_grid	2018_scale_0.50	0.500000	2.033805	0
probit_offset	factor_scale_grid	2018_scale_0.80	0.800000	2.033756	0
probit_offset	utility_scale	2018	2.210128	n/a	0
probit_offset	factor_scale_grid	2019_scale_0.00	0	2.055180	0
probit_offset	factor_scale_grid	2019_scale_0.15	0.150000	2.055151	0
probit_offset	factor_scale_grid	2019_scale_0.30	0.300000	2.055109	0
probit_offset	factor_scale_grid	2019_scale_0.50	0.500000	2.055108	0
probit_offset	factor_scale_grid	2019_scale_0.80	0.800000	2.055190	0

24 rows from the sensitivity artifact. If the probabilities move materially with the integration settings, the result is a property of the integration rather than of the model.

Limitations and the retrospective-odds warning

Retrospective odds, restated

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Probit Offset with a pace factor against the incumbent Offset model on the same margins. Beating a comparator and beating the market are different claims.
- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.